

Chapter 11D. Bullous, Drug, and Severe Cutaneous Reactions

Dermatology: autoimmune blistering disease, dermatitis herpetiformis, erythema multiforme, SJS/TEN, DRESS, AGEP, exfoliative dermatitis, and urgent rash triage.

This chapter bridges pattern recognition with emergency dermatology. Many benign rashes can be diagnosed visually, but dangerous rashes are managed by physiology: mucosal barrier failure, fluid loss, ocular injury, sepsis risk, organ involvement, and medication exposure timing. The core habit is to decide first whether the patient has a benign localized eruption, an autoimmune blistering disease requiring biopsy and immunosuppression, or a severe cutaneous adverse reaction requiring immediate drug cessation and inpatient supportive care.

High-yield: A painful rash with mucosal erosions, skin tenderness, dusky or purpuric macules, rapidly spreading blisters, fever, hypotension, facial edema, eosinophilia, hepatitis, nephritis, ocular symptoms, or altered mental status should be treated as dangerous until proven otherwise.

1. First-pass approach to blisters and severe rashes

A blister is a fluid collection within or beneath the epidermis. The first diagnostic split is whether the blister is fragile and superficial or tense and deeper. Superficial intraepidermal cleavage creates flaccid bullae and erosions that rupture easily; this is the pattern of pemphigus vulgaris. Subepidermal separation creates tense bullae that resist rupture; this is the pattern of bullous pemphigoid, dermatitis herpetiformis, linear IgA disease, porphyria cutanea tarda, and many mechanical or friction blisters.

The second diagnostic split is whether the process is autoimmune, infectious, drug-triggered, inherited, metabolic, or traumatic. The physical exam should document morphology, distribution, mucosal involvement, Nikolsky sign, skin pain versus itch, fever, lymphadenopathy, facial edema, palms/soles involvement, ocular injection, genital erosions, and approximate body surface area affected.

A true Nikolsky sign is lateral shearing of clinically normal-appearing skin producing epidermal detachment. It is classically positive in pemphigus vulgaris and SJS/TEN, and usually negative in bullous pemphigoid. The sign is helpful but not perfectly specific; it should not replace biopsy, medication review, and assessment of systemic toxicity.

High-yield: Blister triage algorithm: 1. Mucosal erosions, fever, skin pain, hypotension, ocular symptoms, or widespread detachment require emergency evaluation. 2. Tense pruritic bullae in an older adult suggest bullous pemphigoid; biopsy perilesional skin for DIF. 3. Flaccid bullae or painful oral erosions suggest pemphigus vulgaris. 4. Grouped intensely pruritic extensor vesicles suggest dermatitis herpetiformis. 5. Tender dusky macules with sloughing after a high-risk drug suggest SJS/TEN.

2. Diagnostic tools for bullous disease

The most useful test in suspected autoimmune blistering disease is skin biopsy, but the biopsy technique matters. For routine H&E histology, sample the edge of a fresh blister so the pathologist can see the level of split and inflammatory infiltrate. For direct immunofluorescence, sample normal-appearing perilesional skin adjacent to a fresh lesion, not necrotic ulcer base.

Direct immunofluorescence often gives the defining pattern: intercellular net-like IgG in pemphigus vulgaris, linear IgG/C3 at the basement membrane in bullous pemphigoid, granular IgA in dermal papillae in dermatitis herpetiformis, and linear IgA at the basement membrane in linear IgA bullous dermatosis.

3. Pemphigus vulgaris

Pemphigus vulgaris is an autoimmune blistering disorder caused by IgG autoantibodies against desmosomal cadherins, especially desmoglein 3 and sometimes desmoglein 1. Desmosomes hold keratinocytes together. When they are disrupted, keratinocytes detach from one another, producing intraepidermal blisters. Because the blister roof is thin, intact blisters may be uncommon; the patient often presents with painful erosions, crusting, and nonhealing oral ulcers.

Mucosal involvement is the clinical clue. Oral erosions may precede skin disease by weeks to months and can cause odynophagia, poor intake, weight loss, and secondary infection. Cutaneous lesions are flaccid bullae or erosions on the scalp, face, trunk, intertriginous areas, and pressure/friction sites. Nikolsky sign is often positive. Untreated pemphigus can be life-threatening due to fluid loss, infection, malnutrition, and complications of high-dose immunosuppression.

High-yield: Pemphigus vulgaris = painful oral erosions + flaccid bullae/erosions + positive Nikolsky + intercellular fishnet IgG on direct immunofluorescence.

4. Bullous pemphigoid

Bullous pemphigoid is the most common autoimmune blistering disease of older adults. Autoantibodies target hemidesmosomal proteins BP180 and BP230, which anchor basal keratinocytes to the basement membrane. Unlike pemphigus, keratinocytes remain attached to each other, but the epidermis separates from the dermis. The result is a subepidermal blister with a thick roof, producing tense bullae that do not rupture easily.

The prodrome is often intensely pruritic urticarial or eczematous plaques before blisters appear. Bullae occur on the trunk, flexural extremities, axillae, groin, and lower abdomen. Mucosal involvement is absent or mild in most cases, and Nikolsky sign is usually negative. Treatment balances disease control against steroid toxicity, infection, skin atrophy, diabetes, osteoporosis, and delirium in older patients.

High-yield: Bullous pemphigoid = older adult + severe itch + tense bullae + negative Nikolsky + linear IgG/C3 along the basement membrane.

5. Dermatitis herpetiformis and celiac-linked blistering disease

Dermatitis herpetiformis is an intensely pruritic autoimmune blistering eruption strongly associated with celiac disease. Despite the name, it is not caused by herpesvirus. IgA antibodies related to gluten-sensitive enteropathy cross-react with epidermal transglutaminase, and granular IgA deposits in dermal papillae trigger neutrophilic microabscesses and subepidermal vesicles.

Patients describe burning or stinging itch and grouped vesicles or excoriated papules on extensor elbows, knees, buttocks, scalp, back, and shoulders. Because the itch is so intense, intact vesicles may be absent. Diagnosis is supported by direct immunofluorescence showing granular IgA in dermal papillae and by celiac serologies such as anti-tissue transglutaminase or anti-endomysial antibodies. Dapsone rapidly improves itch, but strict gluten avoidance treats the underlying disease.

High-yield: Dermatitis herpetiformis = very itchy grouped vesicles on extensor surfaces + gluten-sensitive enteropathy + granular IgA in dermal papillae.

6. Erythema multiforme

Erythema multiforme is an immune-mediated mucocutaneous eruption characterized by target lesions. It is now considered distinct from SJS/TEN rather than a mild point on the same continuum. The most common trigger is infection, especially herpes simplex virus; *Mycoplasma pneumoniae* is another classic association. Drugs can trigger erythema multiforme, but they are much less dominant than in SJS/TEN.

Classic target lesions have three zones: a dusky or vesicular center, a pale edematous ring, and an outer erythematous rim. They favor acral surfaces, especially the dorsal hands, palms, soles, extensor extremities, elbows, knees, and face. Erythema multiforme minor has little or no mucosal involvement. Erythema multiforme major has mucosal involvement but usually lacks the extensive sheet-like epidermal detachment and high mortality risk of SJS/TEN.

7. Stevens-Johnson syndrome and toxic epidermal necrolysis

Stevens-Johnson syndrome and toxic epidermal necrolysis are severe mucocutaneous hypersensitivity reactions with keratinocyte apoptosis and full-thickness epidermal necrosis. SJS and TEN are classified by body surface area of epidermal detachment: SJS involves less than 10%, overlap involves 10-30%, and TEN involves more than 30%. The distinction is prognostic, but early management is similar: stop the culprit drug, admit the patient, protect the airway and mucosa, replace fluids, control pain, prevent infection, and involve dermatology, ophthalmology, and burn or ICU-level teams when appropriate.

The illness often begins with fever, malaise, sore throat, cough, conjunctival burning, and skin pain before the eruption becomes obvious. Lesions are tender dusky macules or atypical flat targets that coalesce. Mucosal involvement is common and may affect oral, ocular, genital, respiratory, and GI surfaces. Ocular injury can lead to corneal ulceration, scarring, and long-term vision loss, so ophthalmology evaluation is mandatory when eye symptoms or conjunctivitis are present.

High-risk drugs include allopurinol, sulfonamide antibiotics, anticonvulsants such as lamotrigine, carbamazepine, phenytoin, and phenobarbital, nevirapine, oxycam NSAIDs, and some antibiotics. Genetic risk matters for selected drugs and populations, including HLA-B*1502 before carbamazepine in many patients of Southeast Asian ancestry and HLA-B*5801 with allopurinol hypersensitivity risk in several populations.

High-yield: SJS/TEN emergency algorithm: 1. Stop all nonessential new drugs immediately. 2. Admit for supportive care and consider ICU/burn unit when detachment is extensive. 3. Consult dermatology and ophthalmology early. 4. Use nonadherent dressings, fluid/electrolyte replacement, nutrition, analgesia, and infection surveillance. 5. Discuss immunomodulation with dermatology; practices vary.

8. DRESS syndrome

DRESS stands for drug reaction with eosinophilia and systemic symptoms. It is a severe delayed drug reaction with rash plus internal organ involvement. The timeline is a major clue: DRESS often begins 2-8 weeks after starting the culprit drug, later than urticaria, fixed drug eruption, or AGEP. Common triggers include aromatic anticonvulsants, allopurinol, sulfonamides, minocycline, dapsone, vancomycin, and other antibiotics.

Clinical findings include fever, diffuse morbilliform eruption, facial edema, lymphadenopathy, eosinophilia or atypical lymphocytosis, and organ involvement. The liver is most commonly affected, but kidneys, lungs, heart, thyroid, pancreas, and CNS may be involved. Myocarditis is an important cause of death and may be clinically subtle, so dyspnea, chest pain, persistent tachycardia, troponin elevation, or ECG abnormalities should prompt urgent evaluation.

High-yield: DRESS is not just a rash with eosinophilia. It is a systemic drug reaction. Check CBC with differential, CMP/LFTs, creatinine, urinalysis, and organ-directed testing when symptoms suggest myocarditis, pneumonitis, nephritis, or hepatitis.

9. Acute generalized exanthematous pustulosis

Acute generalized exanthematous pustulosis, or AGEP, is a severe drug eruption characterized by sudden onset of dozens to hundreds of small sterile nonfollicular pustules on a background of erythema, often beginning in intertriginous areas and spreading rapidly. It is most commonly triggered by antibiotics, especially beta-lactams and macrolides, but many drugs can cause it.

Compared with DRESS, latency is short, often within 1-2 days of exposure if previously sensitized, and the syndrome usually resolves quickly after drug withdrawal. Fever and neutrophilia are common. Mild hepatic or renal abnormalities may occur, but major organ involvement is usually less prominent than in DRESS. Treatment is primarily discontinuation of the offending drug, supportive care, topical corticosteroids, antipyretics, and monitoring.

High-yield: AGEP = abrupt sterile pustules + fever/neutrophilia + short drug latency + rapid improvement after stopping the drug. DRESS = later onset + eosinophilia/facial edema + organ involvement.

10. Fixed drug eruption and common drug exanthems

Most cutaneous drug eruptions are not life-threatening, but they are clinically important because they can be confused with viral exanthems or early severe reactions. A simple morbilliform drug eruption is a symmetric maculopapular rash that typically begins on the trunk and spreads to extremities, often 1-2 weeks after a new medication. Patients may itch but should not have prominent skin pain, mucosal erosions, blistering, facial edema, hypotension, or organ injury.

Fixed drug eruption produces sharply demarcated round or oval erythematous to violaceous patches or plaques that recur at the same site each time the patient is re-exposed to the drug. Lesions may blister and then heal with postinflammatory hyperpigmentation. Common culprits include NSAIDs, acetaminophen, sulfonamides, tetracyclines, barbiturates, and other antimicrobials.

11. Exfoliative dermatitis and erythroderma

Erythroderma is generalized erythema and scaling involving most of the body surface area. It is a dermatologic emergency because the skin is an organ of temperature regulation, fluid balance, and infection defense. When widespread inflammation disrupts barrier function, patients may develop hypothermia or hyperthermia, high-output heart failure, edema, electrolyte abnormalities, dehydration, secondary infection, and protein loss.

Common causes include psoriasis, atopic dermatitis, drug eruptions, pityriasis rubra pilaris, cutaneous T-cell lymphoma/Sézary syndrome, pemphigus foliaceus, and idiopathic disease. Management starts with stabilization: assess vitals, temperature, fluid status, electrolytes, albumin, infection, and cardiac risk. Stop possible offending drugs, avoid irritants, use bland emollients and wet wraps when appropriate, and involve dermatology for biopsy and disease-specific therapy.

High-yield: Erythroderma is not merely a big rash. Think of it as barrier failure with thermoregulation, fluid, infection, and cardiac consequences.

12. Treatment safety: steroids, immunosuppression, and infection

Blistering and inflammatory skin diseases often require corticosteroids or steroid-sparing immunosuppression, but these medications can worsen infection, mask sepsis, destabilize diabetes, trigger psychiatric symptoms, worsen hypertension, accelerate osteoporosis, and impair wound healing. The decision to start systemic steroids should be linked to a working diagnosis and an infection screen appropriate to the patient.

Before prolonged immunosuppression, evaluate vaccination status, hepatitis B/C, TB risk, HIV when indicated, pregnancy status, baseline CBC/CMP, renal function, and medication interactions. For dapsone, check G6PD when appropriate and monitor for hemolysis and methemoglobinemia. For rituximab, consider hepatitis B reactivation risk. For high-dose or prolonged steroids, plan bone protection, glucose monitoring, gastric risk assessment, *Pneumocystis jirovecii* prophylaxis in selected regimens, and a clear taper/follow-up plan.

High-yield comparison tables

Disease	Autoantigen or mechanism	Morphology	Mucosa	Diagnosis	Treatment concept
Pemphigus vulgaris	IgG against desmoglein 3 +/- 1; loss of desmosomal adhesion	Flaccid bullae and painful erosions	Common, often oral first	Suprabasal acantholysis; fishnet IgG/C3; anti-desmoglein antibodies	Systemic steroids, rituximab, steroid-sparing agents, wound care
Bullous pemphigoid	IgG against BP180/BP230 hemidesmosomes	Tense bullae on urticarial plaques, severe itch	Rare or mild	Subepidermal blister with eosinophils; linear IgG/C3 at basement membrane	High-potency topical steroids if localized; systemic or steroid-sparing therapy if extensive
Dermatitis herpetiformis	IgA related to transglutaminase/celiac disease	Grouped excoriated vesicles on extensor surfaces	No classic erosive mucositis	Granular IgA in dermal papillae; anti-tTG/endomysial antibodies	Gluten-free diet; dapsone for rapid symptom control
Linear IgA bullous dermatosis	Linear IgA at basement membrane;	Annular vesicles or tense bullae,	Variable	Linear IgA on DIF	Stop trigger; dapsone or steroids

	can be drug-induced, classically vancomycin	sometimes string-of-pearls pattern			
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Drug rash type	Typical timing	Morphology	Systemic involvement	Key management
Morbilliform exanthem	1-2 weeks after start; sooner on re-exposure	Symmetric macules/papules, trunk then limbs	None or mild fever/itch; no mucosal erosions	Stop nonessential culprit if possible; topical steroids/antihistamines; monitor for red flags
Fixed drug eruption	Hours-days after re-exposure	Well-demarcated violaceous patch/plaque at same site; may blister	Usually absent	Avoid culprit; topical steroids; document clearly
AGEP	Hours-days	Diffuse sterile nonfollicular pustules on erythema	Fever, neutrophilia; organs less common	Stop drug; supportive/topical steroids; monitor labs
DRESS	2-8 weeks	Morbilliform eruption, facial edema, exfoliation possible	Eosinophilia, hepatitis, nephritis, myocarditis, pneumonitis	Stop drug; organ workup; systemic steroids for significant organ disease
SJS/TEN	Often 1-4 weeks	Tender dusky macules, blistering, epidermal necrosis	Mucosal injury, fluid loss, ocular injury, sepsis risk	Stop drug; admit; burn/ICU-level supportive care; dermatology/ophthalmology

Red flag	Why it matters	Examples to consider
Skin pain out of proportion	Necrosis, SJS/TEN, necrotizing infection, ischemia	SJS/TEN, necrotizing fasciitis, calciphylaxis, purpura fulminans
Mucosal erosions or conjunctivitis	Barrier failure and ocular sequelae	SJS/TEN, pemphigus vulgaris, erythema multiforme major, Behcet disease
Dusky, purpuric, necrotic, or nonblanching lesions	Vascular injury, thrombosis, sepsis, vasculitis	Meningococcemia, DIC, vasculitis, warfarin necrosis, SJS/TEN
Bullae or epidermal detachment	Severe autoimmune blistering or epidermal necrolysis	Pemphigus, bullous pemphigoid, SJS/TEN, staphylococcal scalded skin syndrome
Fever, hypotension, confusion, severe malaise	Systemic toxicity	Sepsis, toxic shock, SJS/TEN, DRESS, disseminated infection
Facial edema plus eosinophilia or LFT elevation	Systemic drug reaction	DRESS
Eye pain, photophobia, decreased vision	Sight-threatening ocular disease	SJS/TEN, ocular HSV/VZV, uveitis, ischemia

Medication	Common dermatology use	Major safety points
Topical high-potency corticosteroids	Bullous pemphigoid, eczema flares, psoriasis plaques	Skin atrophy, striae, adrenal suppression if extensive or occlusive; avoid high potency on face/groin unless directed
Systemic corticosteroids	Pemphigus, severe bullous pemphigoid, DRESS organ disease	Hyperglycemia, infection, osteoporosis, psychosis, hypertension, GI risk, adrenal suppression
Rituximab	Pemphigus and refractory autoimmune blistering disease	Infusion reaction, infection, hypogammaglobulinemia, hepatitis B reactivation; vaccination timing matters
Dapsone	Dermatitis herpetiformis, linear IgA disease	Hemolysis, methemoglobinemia, agranulocytosis; check CBC and consider G6PD risk
Cyclosporine	Selected severe psoriasis, SJS/TEN protocols in some centers	Nephrotoxicity, hypertension, drug interactions; monitor creatinine and BP
Mycophenolate/azathioprine/methotrexate	Steroid-sparing agents in autoimmune blistering disease	Cytopenias, hepatotoxicity, infection, teratogenicity; monitor CBC/CMP and interactions

Board-style diagnostic anchors

- Painful oral erosions with fragile flaccid bullae and fishnet immunofluorescence: pemphigus vulgaris.
- Elderly patient with severe pruritus, tense bullae, negative Nikolsky sign, and linear IgG/C3 at the basement membrane: bullous pemphigoid.
- Grouped pruritic vesicles on elbows, knees, or buttocks with celiac symptoms or anti-tTG antibodies: dermatitis herpetiformis.

- Target lesions on hands/feet 1-2 weeks after HSV: erythema multiforme.
- Fever, mucositis, skin pain, dusky macules, and epidermal sloughing after allopurinol, sulfa drug, lamotrigine, or carbamazepine: SJS/TEN.
- Facial edema, morbilliform rash, eosinophilia, hepatitis, and kidney injury weeks after a new anticonvulsant or allopurinol: DRESS.
- Acute fever and hundreds of sterile pustules shortly after a beta-lactam or macrolide: AGEP.
- Recurrent sharply demarcated violaceous lesion at the same site after taking the same medication: fixed drug eruption.
- Generalized erythema and scale over most of the body with hypothermia, edema, electrolyte risk, or cardiac stress: erythroderma.

Graph-RAG extraction map

For graph-RAG extraction, autoimmune blistering disease nodes should connect morphology to adhesion targets, biopsy pattern, immune deposits, mucosal involvement, and treatment class. Severe cutaneous adverse reaction nodes should connect culprit drugs, latency, morphology, organ involvement, emergency disposition, and long-term sequelae. This structure helps distinguish visually similar rashes through causal edges rather than memorized labels alone.

Node	Important edges
Pemphigus vulgaris	Anti-desmoglein IgG -> desmosome dysfunction -> acantholysis -> flaccid bullae; associated with oral erosions; treated with systemic steroids/rituximab.
Bullous pemphigoid	Anti-BP180/BP230 IgG -> hemidesmosome disruption -> subepidermal blister -> tense bullae; treated with potent topical/systemic anti-inflammatory therapy.
Dermatitis herpetiformis	Celiac disease -> IgA transglutaminase antibodies -> granular IgA in dermal papillae -> extensor pruritic vesicles; treated with gluten-free diet/dapsone.
SJS/TEN	High-risk drugs -> keratinocyte apoptosis -> epidermal necrosis -> mucosal erosions/fluid loss/ocular injury; treated by drug cessation and supportive inpatient care.
DRESS	Delayed drug hypersensitivity -> eosinophilia/lymphocyte activation -> hepatitis/nephritis/myocarditis; treated by drug cessation and organ-directed immunosuppression.
AGEP	Drug hypersensitivity -> neutrophilic sterile pustules -> fever/neutrophilia; improves after drug withdrawal.

References and source synthesis

This chapter synthesizes the uploaded dermatology source pages on dermatologic morphology, allergic/drug reactions, viral/fungal/infestation disease, and premalignant lesions with current clinical reference frameworks for autoimmune blistering disease and severe cutaneous adverse reactions. Content is rewritten in original explanatory form for Medvale study and future graph-RAG extraction.

Selected external references used for clinical updating: American Academy of Dermatology guideline pages for dermatologic disease management; CDC shingles and scabies clinical guidance; Merck Manual Professional and DermNet discussions of SJS/TEN and erythema multiforme; peer-reviewed reviews of pemphigus, bullous pemphigoid, and severe cutaneous adverse reactions.